

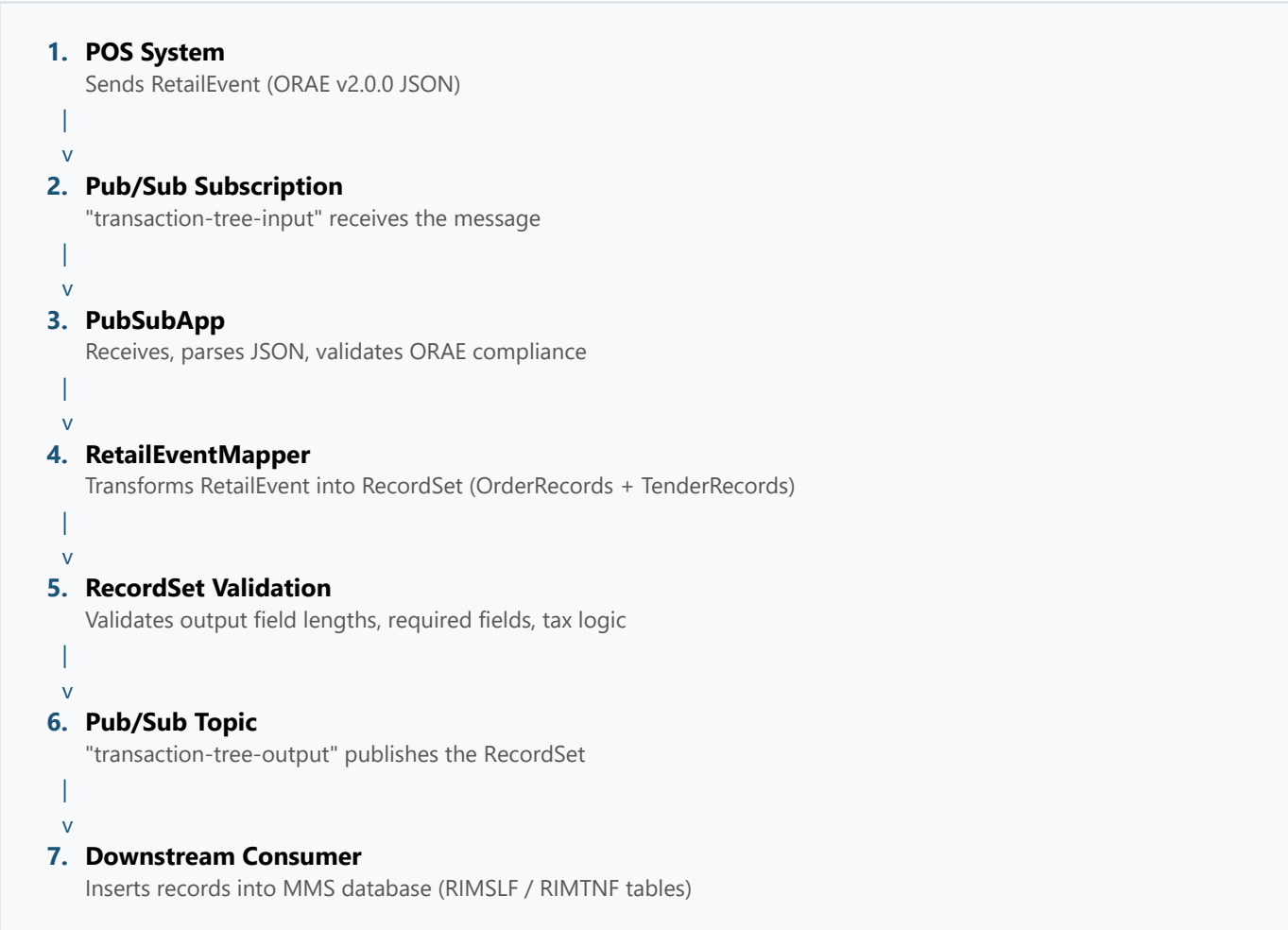
RonaORAEPubSub - System Overview

Retail Transaction Processing via Google Cloud Pub/Sub

1. Purpose

RonaORAEPubSub is a high-throughput Google Cloud Pub/Sub application that receives retail transaction events in ORAE (Open Retail Analytics Event) v2.0.0 format, validates and transforms them into structured database records (OrderRecords for the RIMSLF table and TenderRecords for the RIMTNF table), and publishes the results back to Pub/Sub for downstream database insertion.

2. Architecture & Message Flow



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Logging.cs	File-based logging to c:\opt\transactiontree\pubsub\log\
appsettings.json	Configuration: ProjectId, TopicId, SubscriptionId, paths, feature flags

4. Pub/Sub Configuration

Publisher	Immediate publish, 1 element threshold, 10ms delay
Subscriber	gRPC keepalive (20s ping, 10s timeout), flow control (100 max elements, 10MB), ack extension (30s)
Idle Watchdog	Configurable timeout (default 30 min) with automatic reconnection
Resilience	Automatic subscriber recreation on disconnect, thread-safe async handling

5. Event Types Handled

Event SubTypes:

- transactionSale - Normal retail sale
- transactionBurned - Burned/lost transaction (special handling with audit code)

Transaction Types:

01	SALE	Regular purchase
04	SALE (Emp)	Employee discount purchase
11	RETURN	Item return
43	AR_PAYMENT	Accounts receivable payment
87	VOID	Current void
88	POST_VOID	Post void

6. Error Handling Strategy

Invalid JSON	ACK	Prevent infinite redelivery of bad data
ORAE Validation Failure	ACK	Invalid input will never pass validation
RecordSet Validation Failure	ACK	Malformed output prevented from publishing
Publish Failure	NACK	Retry with exponential backoff (up to 3 attempts)
Auth Failure / High Backlog	ALERT	Slack webhook with 20-min cooldown

7. Data Transformation Highlights

Price & Quantity Formatting:

- Quantities multiplied by 100 (no decimals), padded to 9 digits
- Prices stored without decimals, padded to 9 digits; negative sign flags for compliance

Multi-Province Tax Logic:

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- Ontario (ON): HST 13% consolidated into 1-2 tax records (TaxAuthCode "HON")
- BC/MB/SK/QC: Split federal GST ("FED") + provincial PST/QST per province
- Alberta (AB): Single consolidated GST-only record
- Other provinces: Per-item tax records maintained

Tender / Payment Mapping:

CASH > CA | DEBIT > DC | VISA > VI | MASTERCARD > MA | AMEX > AX | GIFT_CARD > PG | FLEXITI > FX

Other:

- EPP (Extended Protection Plan) tracking via item attributes
- Timezone adjustments based on store region (Western -8h, Eastern -5h)
- SCO (register starts with "8") uses register ID as SalesPerson; ACO uses cashier loginId

8. Environments

Sandbox	prj-n-mime-app-f41w
Development	prj-d-mime-app-tt3x
Production	prj-p-mime-app-h1h6

9. External Integrations

Google Cloud Pub/Sub	Core messaging infrastructure (input subscription + output topic)
Google Cloud Monitoring	Subscription backlog monitoring
Slack	Real-time alerts for auth failures and high backlog (20-min cooldown)
MMS Database	Downstream target via RIMSLF (orders) and RIMTNF (tenders) tables